

Assignment 2

Jason Huang

1. Question 1

- a. Only the pairwise matrix can come back not PSD. Each entry uses different days, so the entries don't have to be consistent with each other. The complete case matrix is one ordinary correlation matrix from the same 81 rows, so $x'Rx$ is a real portfolio variance and can't go negative. Only 81 of 250 days have all five series.

```
Days each pair is jointly observed (diagonal = days each series trades):
  A    B    C    D  IDX
A  242 242 221 93 242
B  242 242 221 93 242
C  221 221 228 83 228
D   93  93  83 96  96
IDX 242 242 228 96 250

Days all five series trade (complete rows): 81 of 250
```

- b. IDX is almost $0.4A + 0.3B + 0.2C + 0.1D$, so long IDX / short the basket has almost no variance. That puts the smallest eigenvalue of the true matrix very close to zero, right on the edge of PSD, so a little noise from using different days can push it negative. This data set is very exposed to that.
- c. Anything with D. D overlaps with the others for only 83 to 96 days vs 221 to 242 for the rest, so its standard errors are about 0.07 to 0.11 vs about 0.05. IDX with A should be the most reliable (about 0.016).

```
Approx. standard error of each pairwise correlation, (1 - rho^2)/sqrt(n - 1):
  A    B    C    D  IDX
A   NaN 0.0503 0.0549 0.0842 0.0157
B  0.0503   NaN 0.0530 0.0963 0.0298
C  0.0549 0.0530   NaN 0.1067 0.0326
D  0.0842 0.0963 0.1067   NaN 0.0650
IDX 0.0157 0.0298 0.0326 0.0650   NaN

Complete case uses only 81 rows, so its standard errors are about (1 - rho^2)/sqrt(80) = e.
g. 0.094 at rho=0.4
```

- d. Complete case eigenvalues are all positive (smallest 0.0015) and Cholesky worked. Pairwise has a negative eigenvalue, -0.0111, and Cholesky failed.

```

Complete case correlation matrix:
      A      B      C      D      IDX
A  1.0000  0.3967  0.2466  0.4136  0.8714
B  0.3967  1.0000  0.3169  0.2361  0.6709
C  0.2466  0.3169  1.0000  0.1852  0.5648
D  0.4136  0.2361  0.1852  1.0000  0.5719
IDX 0.8714  0.6709  0.5648  0.5719  1.0000

Pairwise correlation matrix:
      A      B      C      D      IDX
A  1.0000  0.4686  0.4309  0.4384  0.8696
B  0.4686  1.0000  0.4630  0.2767  0.7329
C  0.4309  0.4630  1.0000  0.1846  0.7136
D  0.4384  0.2767  0.1846  1.0000  0.6057
IDX 0.8696  0.7329  0.7136  0.6057  1.0000

Eigenvalues, complete case: [0.0015  0.5483  0.6933  0.8704  2.8864]
Eigenvalues, pairwise      : [-0.0111  0.4714  0.5284  0.8574  3.1539]
Cholesky of complete case matrix: numpy SUCCEEDED
course chol_psd SUCCEEDED, max |L L' - M| = 1.11e-16
Cholesky of pairwise matrix: numpy FAILED (Matrix is not positive definite)
course chol_psd FAILED (matrix is not positive semi-definite)

```

- e. Pairwise correlations with the full history standard deviations give a tracking portfolio variance of $-1.7994\text{e-}06$, which is negative. Complete case gives $+1.0975\text{e-}06$. The actual variance of the tracking returns on the 81 complete days is $3.22\text{e-}07$.

```

Full history standard deviations: {'A': 0.016502, 'B': 0.011919, 'C': 0.018324, 'D': 0.0228
29, 'IDX': 0.012281}

Tracking portfolio variance, pairwise matrix      : -1.7994e-06 (std = -0.00134)
Tracking portfolio variance, complete case matrix :  1.0975e-06 (std = 0.00105)

```

- f. Both repairs get the smallest eigenvalue to about zero. Frobenius distance from pairwise is 0.01618 for Rebonato and Jackel and 0.01552 for Higham. Tracking variance becomes $6.81\text{e-}07$ and $6.79\text{e-}07$. The strict numpy Cholesky still fails on them because they're exactly singular, but the course Cholesky works.

```

Repairs of the pairwise matrix:
  Rebonato-Jackel: min eigenvalue = -7.515e-17, Frobenius distance from pairwise = 0.0161
8, tracking variance =  6.8073e-07
Cholesky of Rebonato-Jackel matrix: numpy FAILED (Matrix is not positive definite)
course chol_psd SUCCEEDED, max |L L' - M| = 2.22e-16
  Higham: min eigenvalue = -1.443e-11, Frobenius distance from pairwise = 0.0155
2, tracking variance =  6.7876e-07
Cholesky of Higham matrix: numpy FAILED (Matrix is not positive definite)
course chol_psd SUCCEEDED, max |L L' - M| = 2.13e-11

```

- g. PSD means $x'Sx$ is at least 0 for every x , so every portfolio has non-negative variance. The tracking portfolio is basically the eigenvector for the smallest eigenvalue (about -0.39, -0.23, -0.27, -0.21, 0.82), so it lands right on the negative direction and gets a negative variance. A variance can't be negative, so the matrix isn't a valid covariance matrix.
- h. Higham moved the IDX entries the most: A-IDX by -0.0070, C-IDX -0.0048, B-IDX -0.0042, D-IDX -0.0037. Everything else moved about 0.001 to 0.002. In c I guessed D, so I was wrong about where it would show up. The repair doesn't know how many days are behind each entry. It removes the negative eigen-direction, so entries move in proportion to how much they load on that eigenvector, and IDX (0.82) and A (0.39) load the most.

```

Higham change (repaired - pairwise):
      A      B      C      D      IDX
A  0.0000  0.0020  0.0023  0.0018 -0.0070
B  0.0020  0.0000  0.0014  0.0011 -0.0042
C  0.0023  0.0014  0.0000  0.0012 -0.0048
D  0.0018  0.0011  0.0012  0.0000 -0.0037
IDX -0.0070 -0.0042 -0.0048 -0.0037  0.0000
Entries ranked by size of Higham change:
A-IDX: pairwise= 0.8696 Higham= 0.8625 change=-0.0070 (days observed together: 242)
C-IDX: pairwise= 0.7136 Higham= 0.7088 change=-0.0048 (days observed together: 228)
B-IDX: pairwise= 0.7329 Higham= 0.7287 change=-0.0042 (days observed together: 242)
D-IDX: pairwise= 0.6057 Higham= 0.6020 change=-0.0037 (days observed together: 96)
A-C: pairwise= 0.4309 Higham= 0.4332 change= 0.0023 (days observed together: 221)

```

- i. The gap between complete case and pairwise (Frobenius 0.424) is about 27 times bigger than the Higham repair (0.0155). Entries differ by up to about 0.18 between estimators but the repair never moves one by more than 0.01. So the estimator choice matters much more than the repair choice here.

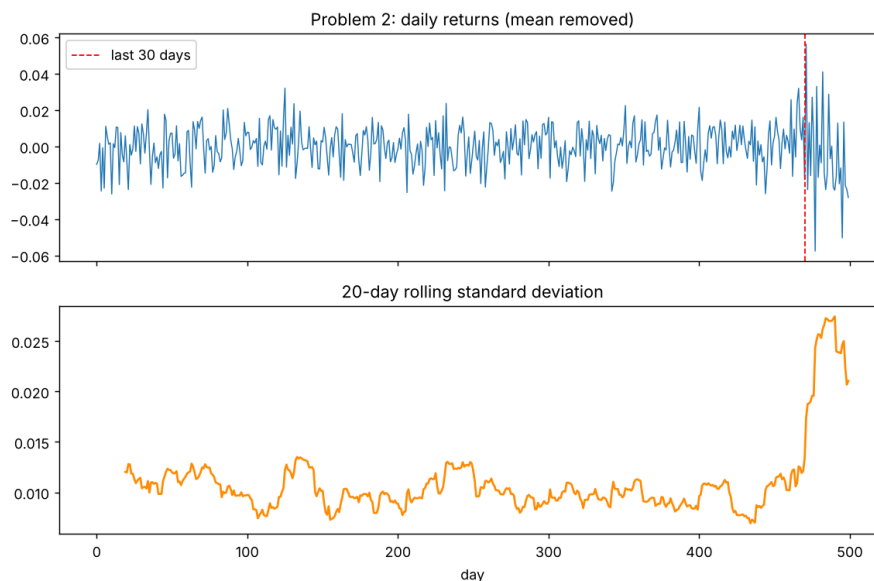
```

Frobenius gap between complete case and pairwise : 0.42393
Frobenius size of Higham repair                  : 0.01552
Frobenius size of Rebonato-Jackel repair         : 0.01618
Tracking variance: complete=1.0975e-06, pairwise=-1.7994e-06, Higham=6.7876e-07, R-J=6.8073e-07

Actual variance of the tracking portfolio's return on the 81 complete days (direct, no correlation matrix): 3.2221e-07

```

2. Question 2



- . The returns look calm for about the first 470 days (around 1% a day), then volatility jumps in the last 30 days or so (rolling 20 day std goes from about 1% to about 2.7%).
- a. My ranking, smallest to largest: t, equal weight normal, historical, EW 0.97, EW 0.94. Equal weight only gives the last 30 days 6% of the weight so it's mostly the calm regime. The t is fit to the same data so it should sit on the calm days and treat the wild ones as tails, a bit below the

normal because of its thinner shoulders. Historical should be close to equal weight. The EW ones load up on the wild days, 0.94 more than 0.97.

- b. Taking the last 30 days as the different ones: lambda 0.94 has an effective sample size of 32.3, half life 11.2 days, and 84.4% of its weight on those days. Lambda 0.97 has 65.7, 22.8 days and 59.9%. Equal weight has 6%.

```
Regime split used (chosen by eye from the plot): last 30 days vs. the first 470

lambda | n_eff (n=500 weights) | n_eff closed form (1+l)/(1-l) | half life (days) | weight
on last 30 days | std error of vol (1/sqrt(2 n_eff))
0.94 | 32.33 | 32.33 | 11.20 |
0.8437 | 0.1244
0.97 | 65.67 | 65.67 | 22.76 |
0.5990 | 0.0873
Equal weight: the last 30 days carry 0.0600 of the weight.
```

- c. Mean 0, std 1.19%, skewness -0.156, excess kurtosis 2.13. I don't think one normal regime can do this. With 500 observations the standard error of kurtosis is about 0.22, so 2.13 is about 10 standard errors from 0. The skewness is only 1.4 standard errors out, so that part is fine.

```
First four moments of the full sample:
returns: mean=-0.000000 var=0.00014161 std=0.011900 skew=-0.1556 excess kurt= 2.1324
Under a normal, the std error of sample skewness is about sqrt(6/n) = 0.110 and of excess k
urtosis sqrt(24/n) = 0.219
-> skewness is -1.4 standard errors from 0, excess kurtosis is 9.7 standard errors from 0
```

- d. VaR on \$1,000,000: equal weight \$19,594, EW 0.97 \$34,619, EW 0.94 \$38,707, t \$19,057, historical \$20,574. The fitted t has scale 0.010306 and nu = 8.36.

```
Daily std: equal weight=0.01191, EW 0.97=0.02105, EW 0.94=0.02353
Fitted t: location=-0.000000, scale=0.010306, nu=8.3606 (t std = scale*sqrt(nu/(nu-2)) =
0.01182)

5% one day VaR on $1,000,000:
1. Normal, equally weighted std $ 19,594
2. Normal, EW std lambda=0.97 $ 34,619
3. Normal, EW std lambda=0.94 $ 38,707
4. Student t (MLE) $ 19,057
5. Historical simulation $ 20,574
Ranking, smallest to largest: 4 < 1 < 5 < 2 < 3
```

- e. The order is t, equal weight, historical, EW 0.97, EW 0.94, so I got both ends right. The one miss is historical sitting a bit above equal weight normal (20,574 vs 19,594). The worst days (about -5.7% and -5.0%) are in the new regime and land in the left tail, and the returns are slightly left skewed, which the normal can't see.
- f. Std is 1.04% in the first 470 days and 2.56% in the last 30, about 2.5 times higher. Each regime on its own looks normal (excess kurtosis -0.12 and 0.13). Mixing two normals with those stds at 94% and 6% gives excess kurtosis 2.57, close to the 2.13 I measured. So the fat tails come from mixing two regimes, and the fitted t (nu 8.4) is describing that blend. That's why its VaR is basically the calm regime number, while the recent regime alone would give about \$42k.

```

Std in old regime (first 470 days): 0.01038   Std in recent regime (last 30 days): 0.02559
ratio = 2.47
old: mean= 0.000406   var=0.00010749   std=0.010368   skew=-0.1399   excess kurt=-0.1178
recent: mean=-0.006364   var=0.00063317   std=0.025163   skew= 0.5251   excess kurt= 0.1268
Kurtosis of a 2-regime normal mixture with these stds and weights: 5.565 (excess 2.565); sa
mple excess kurtosis = 2.132
t with nu=8.36 implies excess kurtosis 6/(nu-4) = 1.376
VaR if we used each regime's own normal: old=$17,071, recent=$42,097

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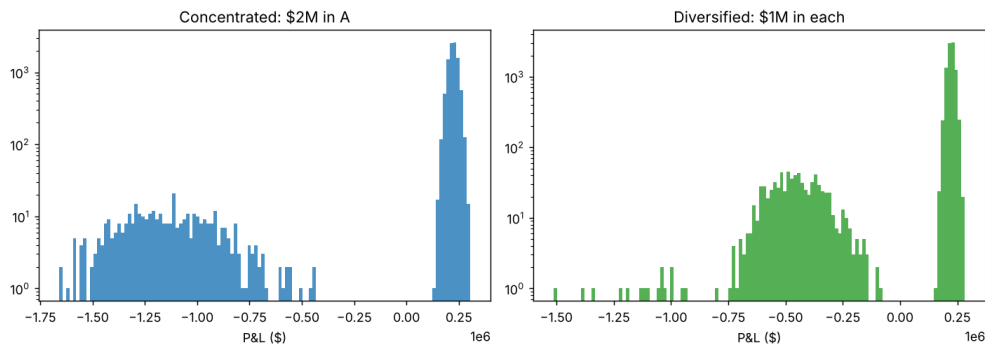
- g. The EW 0.94 vol is 11.8% above the 0.97 vol. The standard error is about 12.4% at 0.94 and 8.7% at 0.97, so the gap is only about one standard error, within the noise (the two share data, so their errors move together and it's a bit better than that). Both are way above equal weight though (about +98%), so the regime change is real. My opinion: 0.94 for a trading limit since it reacts fast, 0.97 for a capital number since it's more stable.

```

Relative gap between the two EW vols (0.94 vs 0.97): 0.1181
noise (std error / vol) at lambda=0.94: 0.1244
noise (std error / vol) at lambda=0.97: 0.0873
EW 0.94 vol vs equal weight vol, relative gap: 0.9755
Dollar VaR gap 0.94 vs 0.97: $4,088

```

3. Question 3



- a. Both bonds are very skewed (about -4.8 and -4.9) with huge excess kurtosis (about 22 and 23). Each one makes about +11.1% almost every time and has a rare big default loss. A normal VaR is a symmetric bell around the mean, so it can't see that and I expect it to be way off.

```

10000 scenarios; each bond bought at 90, pays 100 if no default -> no-default return = 0.11
11

First four moments of each bond's return:
A: mean= 0.083709   var=0.01849080   std=0.135981   skew=-4.8225   excess kurt= 22.0820
B: mean= 0.085011   var=0.01744913   std=0.132095   skew=-4.9148   excess kurt= 23.0744

```

- b. A loses more than 20% in 406 scenarios, B in 393, both in only 17, and at least one in 782 (7.82%). Each bond alone has under a 5% chance of default, so its 5% VaR should sit above the default zone, around zero or even a gain. The pair has a 7.8% chance that one defaults, so its 5% cutoff lands in the default losses. I predict the diversified VaR is larger. For ES I predict the diversified one is smaller, since each default only hits half the money.

```

Scenarios where A loses more than 20%: 406 (4.06%)
Scenarios where B loses more than 20%: 393 (3.93%)
Scenarios where BOTH lose more than 20%: 17
Scenarios where AT LEAST ONE loses more than 20%: 782 (7.82%)
Worst single return: A=-0.8297, B=-0.8560; scenarios worse than -50%: A=285, B=258
Share of A scenarios with a gain (return > 0): 0.9594
Correlation of the two bonds' returns: 0.0022

```

- c. Historical 5%: \$1M in A has VaR -84,531 and ES 443,628. \$1M in B has -86,011 and 419,587. \$2M in A has -169,061 and 887,257. \$1M in each has 414,767 and 541,997. The negative VaRs are real, at the 5% point the single bond is still making money.

```

--- alpha = 5% ---
Position      Hist VaR      Hist ES      Normal VaR      Normal ES
$1M in A      -84,531      443,628      139,971      196,794
$1M in B      -86,011      419,587      132,277      187,477
$2M in A      -169,061     887,257      279,941      393,589
$1M A + $1M B  414,767     541,997      143,466      222,774

--- alpha = 1% ---
Position      Hist VaR      Hist ES      Normal VaR      Normal ES
$1M in A      650,545     709,548     232,645     278,727
$1M in B      641,248     700,805     222,303     267,068
$2M in A     1,301,090   1,419,096     465,291     557,454
$1M A + $1M B  598,237     716,408     272,811     337,127

```

- d. Normal VaR at 5%: 139,971 for A, 132,277 for B, 279,941 for \$2M in A, 143,466 for \$1M in each (in the screenshot above).
- e. At 1% (same screenshot) historical VaR is 650,545, 641,248, 1,301,090 and 598,237, and ES is 709,548, 700,805, 1,419,096 and 716,408.
- f. At 5% $VaR(A) + VaR(B) = -170,542$ but VaR of the combination is 414,767, so VaR is not subadditive here. ES is: 863,216 for the two separately vs 541,997 combined. Each bond alone has a default chance under 5% so VaR never sees a default, but the pair has a 7.8% chance of at least one, so the cutoff now falls in the default losses. VaR only looks at one point and ignores how bad it gets after it. ES averages the tail so it sees the benefit.

```

Subadditivity check at 5% (historical):
  VaR: A + B separately = -170,542; combined = 414,767; combined - sum = 585,309 -> VIOLATED
  ES: A + B separately = 863,216; combined = 541,997; combined - sum = -321,218 -> SUBADDITIVE
Subadditivity check at 1% (historical):
  VaR: A + B separately = 1,291,793; combined = 598,237; combined - sum = -693,556 -> SUBADDITIVE
  ES: A + B separately = 1,410,353; combined = 716,408; combined - sum = -693,945 -> SUBADDITIVE

Probability A defaults (loss > 20%) = 0.0406; P(at least one of two defaults) = 0.0782
5th percentile of A return = 0.0846; 5th percentile of the average of A and B = -0.2073

```

- g. Normal VaR favors the diversified choice (143k vs 280k), which is the right choice (ES agrees, 542k vs 887k), but for the wrong reason. It's just the square root of 2 effect from adding two independent things and knows nothing about defaults. Its numbers are far off: 143k for the diversified position when the historical number is 415k, and 280k for the concentrated one when the historical number is a gain.

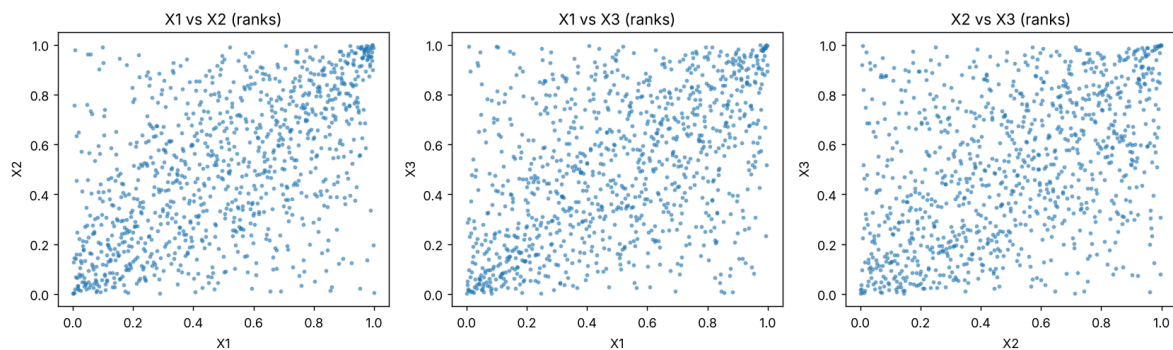
- h. At 1% $\text{VaR}(A) + \text{VaR}(B) = 1,291,793$ vs $598,237$ combined, so VaR is subadditive and diversification looks good. The 1% cutoff is inside the default zone for each bond alone as well as for the pair, so VaR sees a default in every case and the smaller loss per default shows up. So when VaR fails depends on where the confidence level sits relative to the default probability. If the tail probability is just under the chance of the bad event (5% here), VaR is blind to it, and a small change in alpha changes the answer.

4. Question 4

- a. From the moments I'd fit a normal to X1 (skew and excess kurtosis about 0) and a t to X2 and X3 (excess kurtosis 7.08 and 2.85). X2's 7.08 looks out of place, so I checked: row 951 is a -14.3% day for X2 (8.8 standard deviations) and it's also the worst day for X1 and X3. Without it X2's excess kurtosis drops to 1.71 and its skew to +0.06 (X3's goes to 1.72). So X2's number leans on one day, but both X2 and X3 still have some excess kurtosis without it, so I'd keep the t for both.

```
First four moments:
X1: mean= 0.000257 var=0.00014133 std=0.011888 skew=-0.0342 excess kurt=-0.1225
X2: mean=-0.000252 var=0.00026116 std=0.016160 skew=-0.6141 excess kurt= 7.0834
X3: mean= 0.000890 var=0.00010028 std=0.010014 skew= 0.0740 excess kurt= 2.8537

Sensitivity: recompute after dropping the single most extreme observation of each series
X1: dropped row 951 (value -0.0400, 3.4 std devs out)
X1 w/o: mean= 0.000297 var=0.00013985 std=0.011826 skew=-0.0055 excess kurt=-0.1927
X2: dropped row 951 (value -0.1429, 8.8 std devs out)
X2 w/o: mean=-0.000110 var=0.00024103 std=0.015525 skew= 0.0558 excess kurt= 1.7101
X3: dropped row 951 (value -0.0612, 6.2 std devs out)
X3 w/o: mean= 0.000952 var=0.00009651 std=0.009824 skew= 0.3127 excess kurt= 1.7161
```



- b. Under independence I'd expect $1000 \times 0.025 \times 0.025 = 0.625$ days. The data has 6 / 7 / 6 joint worst days and 8 / 12 / 9 joint best days for X1-X2 / X1-X3 / X2-X3, roughly 10 to 20 times more. The plots show tight corners at the lower left and upper right. Some of that is just correlation (Kendall's tau 0.31 to 0.40), but the corners still look more connected than the middle. I predict the t copula wins.

```
Joint tail days: both series in their own worst 2.5% / best 2.5%
Expected under independence over 1000 days: 0.625
X1-X2: both worst = 6, both best = 8 (Kendall tau = 0.3964)
X1-X3: both worst = 7, both best = 12 (Kendall tau = 0.3354)
X2-X3: both worst = 6, both best = 9 (Kendall tau = 0.3074)
```

- c. AICc picks normal for X1 (-6022.54 vs -6020.52, the t's nu runs off to infinity), t for X2 (-5518.18 vs -5408.49, nu 5.27) and t for X3 (-6428.02 vs -6365.69, nu 6.09). Same as my prediction.

```
Margin fits (AICc, lower is better):
X1: Normal AICc= -6022.54   t AICc= -6020.52   -> Normal   (t: nu=61111333659251.80, scale=0.01189)
X2: Normal AICc= -5408.49   t AICc= -5518.18   -> t         (t: nu=5.27, scale=0.01254)
X3: Normal AICc= -6365.69   t AICc= -6428.02   -> t         (t: nu=6.09, scale=0.00818)
```

- d. R comes from Kendall's tau and is the same for both copulas, with margins frozen, so the Gaussian has 0 free parameters and the t has 1 (nu). Gaussian: log likelihood 350.454, AICc -700.907, BIC -700.907. t: log likelihood 451.999, nu 3.658, AICc -901.994, BIC -897.090.

```
R from Kendall's tau:
      X1      X2      X3
X1 1.0000 0.5833 0.5029
X2 0.5833 1.0000 0.4644
X3 0.5029 0.4644 1.0000
Eigenvalues of R: [0.4128 0.5519 2.0353]

      logL      nu      AICc      BIC
Gaussian  350.454    inf  -700.907  -700.907
t         451.999   3.658  -901.994  -897.090
logL difference (t - Gauss) = 101.545; delta BIC = 2*dlogL - ln(m) = 196.183; delta AICc
(Gauss - t) = 201.086
```

- e. VaR and ES for \$1M in each asset, 100,000 simulated days per copula, and historical:

```
Portfolio ($1M in each of X1-X3) VaR / ES, 100,000 simulated days:
      VaR 5%      ES 5%      VaR 1%      ES 1%
Gaussian  49,420    66,393    76,251    93,840
t copula  48,882    68,372    79,628    102,806
Historical 47,939    65,699    70,092    102,424
t vs Gauss % -1.1%      3.0%      4.4%      9.6%
```

- f. Joint tail days per 1000 days implied by each copula (the data counts are in the b screenshot):

```
Implied joint tail days per 1000 days (simulated copula uniforms, tail = 2.5%):
pair data lo data hi Gauss lo Gauss hi t lo t hi
X1-X2 6 8 6.39 6.38 9.62 9.97
X1-X3 7 12 4.61 4.68 8.11 8.49
X2-X3 6 9 4.55 4.49 7.74 7.92
```

- g. The t copula wins by a lot. Log likelihood is 101.5 higher, delta BIC is $2(101.5) - \ln(1000) = 196$ and delta AICc is 201. Nu is 3.66, a very heavy tailed copula. The Gaussian under-counts joint tail days in 5 of the 6 cells (X1-X3 best: 4.7 vs 12 in the data). The t is closer overall (total absolute miss about 13.0 vs 17.7 for the Gaussian), but it's a bit too high on the lower corners (9.6 vs 6 for X1-X2), and the data has more upper joint tail days than lower (29 vs 19), which a symmetric t copula can't capture.
- h. The 1% ES moves the most (+9.6%, 93,840 to 102,806) and the 5% VaR barely moves (-1.1%). Both copulas have the same margins and correlation matrix, so they agree on the body of the distribution, and the 5% VaR sits close enough to the body that they look alike. The t only differs far out where the assets crash together, which is exactly what the 1% ES averages over. The historical 1% ES (102,424) is close to the t, though it's only an average of 10 days.
- i. Total log likelihood gap is 101.5. The 793 days where no series is in its outer 5% contribute 50.5 of it, which is 49.7%. So about half of the t's advantage comes from ordinary days. The 68 days with two or more series in an outer 5% together give 68.5 on their own, and the biggest single

day adds 10.3 (probably row 951). So the information criterion isn't scoring only the tails, it scores how well the whole joint density fits all 1000 days, and a big BIC gap doesn't prove by itself that the tail dependence is real.

```
Total ell_t - ell_Gauss = 101.545
Days with no series in its outer 5% on either side: 793 of 1000 (79.3%)
  their contribution: 50.518 (49.7% of the total)
Days with at least one series in its outer 5%: 207; contribution 51.027 (50.3%)
Days with 2+ series in an outer 5%: 68; contribution 68.526 (67.5%)
Largest single day contribution: 10.289; smallest: -1.761; days where t beats Gauss: 579
```

- j. The most correlated pair is X1-X2 (rho 0.5833). Under the fitted t copula (nu 3.66) lower tail dependence is 0.322, so given one is deep in its tail there's about a 32% chance the other is too. Under the Gaussian it's 0. At the same rho the t would give 0.12 at nu 10 and 0.03 at nu 20, so it depends a lot on nu.

```
Most correlated pair: X1-X2 with rho = 0.5833
t copula (nu=3.66) lower tail dependence = 0.3220
```

5. Question 5

- a. $\text{Var}(P) = w_A^2 \text{Var}(A) + w_B^2 \text{Var}(B) + 2 w_A w_B \text{Cov}(A, B)$, and $\text{Cov}(A, B) = b_A b_B \text{var}(M) + \text{Cov}(e_A, e_B)$. If the stocks share an industry the market misses, the error covariance is positive. For P1 (long both) that adds variance, so setting it to zero understates P1's VaR. For P2 (long A, short B) it acts like a hedge, so ignoring it overstates P2's VaR.
- b. A: alpha 0.000205, beta 1.0353, residual std 0.012901. B: alpha 0.000315, beta 0.9354, residual std 0.011550. The residual correlation is 0.698, so they share something the market doesn't explain.

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Stock A: alpha= 0.000205 beta=1.0353 residual std=0.012901 R^2=0.4181
Stock B: alpha= 0.000315 beta=0.9354 residual std=0.011550 R^2=0.4226
Correlation between the residuals: 0.6983
Market return std: 0.010553
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- c. From 100,000 simulated days (alpha set to 0 for zero expected returns, same random numbers in both runs), 5% VaR for P1 is \$50,441 with the full residual covariance vs \$44,637 with the off diagonal at zero (-11.5%). For P2 it's \$15,868 vs \$28,640 (+80.5%).

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Simulated 5% VaR (100,000 days), zero expected returns:
                                full residual cov  diag residual cov  diag / full - 1
P1 (long A, long B)              50,441             44,637             -11.5%
P2 (long A, short B)             15,868             28,640             80.5%
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- d. Delta normal VaR from the sample covariance: P1 \$50,415, P2 \$15,834.

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Delta normal VaR from the sample covariance (zero mean):
P1 (long A, long B): $50,415 | simulated full residual cov VaR: $50,441 | difference:
0.05%
P2 (long A, short B): $15,834 | simulated full residual cov VaR: $15,868 | difference:
0.22%
Correlation between the two stock returns: 0.8251
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- e. Yes, both moved the way I predicted. The shortcut understates P1 by 11.5% and overstates P2 by 80.5%. P2 is much more exposed because its market exposure almost cancels (beta 1.04

long vs 0.94 short), so its variance is almost all residual: about 91.6M with the full covariance vs 299M with the shortcut, which throws away a cross term of about -208M. For P1 the market part (432M) and residual part (507M) are both big, so the lost +208M is a smaller piece.

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Portfolio standard deviations from the Week 2 formula  $w' \Sigma w$  ( $\Sigma = s^2 M \beta \beta' + \Sigma_{\epsilon}$ ):
P1 (long A, long B): full=30,650 diag=27,050 -> normal VaR full=50,415, diag=44,494
variance split: market part=432,481,449, residual part (full)=506,935,668, residual part (diag)=299,243,934; cross term lost by the shortcut = 207,691,734
P2 (long A, short B): full=9,626 diag=17,331 -> normal VaR full=15,834, diag=28,507
variance split: market part=1,110,499, residual part (full)=91,552,200, residual part (diag)=299,243,934; cross term lost by the shortcut = -207,691,734
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- f. The full residual covariance VaRs match the delta normal ones to within 0.05% and 0.22%, which is simulation noise. OLS is a linear function of normals, so the market and errors together are a joint normal and the regression is already inside the joint normal. The covariance the model implies is the same as the sample covariance of the two stocks (I printed both), so for a linear portfolio it's the same calculation. They'd stop agreeing if the positions were nonlinear (options), if the market or errors weren't normal, if the errors depend on the market even though they're uncorrelated with it (volatility rising when the market falls), or if the residual covariance is changed like in c.